

Penuel Stanley-Zebulon

Penn State University, Harrisburg, PA | (717) 686-6657 | stanleyzebulonp@gmail.com | linkedin.com/in/penuel-stanley-zebulon

EDUCATION

Pennsylvania State University - Harrisburg, PA

B.S. in Artificial Intelligence Methods and Applications (AIMA)

Expected: May 2028

Pan-Atlantic University - Lagos, Nigeria

Bachelor's coursework in Electrical/Electronics Engineering

Nov. 2022 - Jul. 2023

Relevant Coursework: Artificial Intelligence, Statistics, Math for Machine Learning, Object-Oriented Programming, Data Structures and Algorithms, Discrete Math for Computer Science

Certification: SAP Certified - SAP Generative AI Developer (100% system-based assessment), Jul. 2026

SELECTED AI SYSTEMS & PRODUCT PROJECTS

Alethia Live - Real-Time Healthcare Navigation Assistant

Mar. 2026

[GitHub](#) | [Live Demo](#) | *Technologies: Next.js, TypeScript, Gemini Live API, WebSockets, Google Cloud Run*

- Built a real-time voice and document-guidance assistant that helps users understand after-visit summaries and discharge instructions through live microphone input, streamed audio, and screenshot explanation.
- Implemented ephemeral-token authentication, WebSocket setup, structured outputs, red-flag awareness, and guardrails that prevent diagnosis or treatment advice.

Sema - Patient Evidence Packet Agent

Jun. 2026

[GitHub](#) | [Live Demo](#) | *Technologies: Python, React, TypeScript, Tailwind CSS, automated testing, browser PDF export*

- Built a multimodal healthcare AI agent that helps patients organize stories, audio, images, documents, and movement observations into a structured packet for clinical conversations.
- Designed review states, permission controls, schema validation, and deterministic fallbacks, then validated safety, voice, and packet workflows with 217 automated checks.

Neonatal Brain MRI Developmental Age-Bin Classifier

Dec. 2025 - Jan. 2026

[GitHub](#) | [Live Demo](#) | *Technologies: Python, EfficientNet-B0, transfer learning, Grad-CAM, scikit-learn*

- Achieved 87.8% macro-F1 and 87.7% accuracy across 29,516 neonatal and infant MRI slices and 14 developmental-age classes by fine-tuning EfficientNet-B0 with transfer learning.
- Reduced near-duplicate leakage risk through perceptual-hash grouping and group-based splits, then evaluated per-class F1, confusion matrices, Grad-CAM visualizations, and reproducible artifacts.

WORK EXPERIENCE

IBM AI Experiential Lab - Healthcare Track

Feb. 2026 - May 2026

Healthcare AI Product Builder

- Reduced simulated referral-preparation time by approximately 40% by building Mamathemba, a full-stack maternal-health referral-readiness prototype with structured case intake, missing-information checks, facility comparison, visible destination rationale, and editable handoff notes.
- Implemented the workflow with Next.js, FastAPI, and Python while keeping clinicians responsible for the final referral decision through human review.

AI4ALL

Mar. 2026

Ignite Fellow - Applied AI in Healthcare

- Achieved approximately 90% test accuracy and 88% seizure recall by training and evaluating a 1D CNN on 9,500+ CHB-MIT multichannel EEG windows for seizure vs. non-seizure classification.
- Shipped a reproducible Google Colab and Streamlit demo with GitHub documentation and presented model performance, limitations, and healthcare AI risks at a national AI4ALL virtual symposium.

Penn State Harrisburg Center for Teaching Excellence

Oct. 2024 - Present

Student Media Production Assistant

- Produced accessible instructional media across 100+ faculty/course videos through captioning, audio descriptions, video editing, and course-media preparation with multimedia staff.

SKILLS

- Programming and Data:** Python, NumPy, pandas, SQL basics, Google Colab, Jupyter
- AI/ML:** CNNs, transfer learning, model evaluation, leakage-aware splits, Grad-CAM, scikit-learn, TensorFlow/Keras, PyTorch fundamentals
- GenAI and Agents:** Gemini Live API, SAP generative AI hub, LLM workflows, prompt engineering, RAG fundamentals, AI agents, agent/tool workflows, structured outputs, safety evaluation, automated testing, guardrails, human-in-the-loop design
- Web and Cloud:** Next.js, React, TypeScript, Tailwind CSS, FastAPI, REST APIs, Streamlit, Git/GitHub, Google Cloud Run

LEADERSHIP & INVOLVEMENT

Student Ambassador - Adobe (Penn State University)

Jan. 2024 - May 2025

- Planned and delivered Adobe Express workshops for Penn State students, teaching creative workflows, project development, and communication skills while improving materials through feedback-driven iteration.

Bible Study Leader - InterVarsity Christian Fellowship/USA

Mar. 2024 - Present

- Lead weekly Bible study discussions, coordinate communication and outreach, and support an inclusive student community focused on belonging, reflection, and faith-based growth.